

WADF105 – Practical Laboratory 2

OPNsense Firewall Policy Testing, Logging and Packet Analysis

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Laboratory: 2

Environment: VMware Workstation – Authorised ICDFIA Laboratory

Date: 5 October 2026

1. Introduction

This laboratory examined OPNsense firewall behaviour by creating two narrow temporary blocking rules, testing their effects from the Ubuntu client, examining firewall logs, correlating the results with Wireshark packet captures, and restoring normal connectivity.

2. Objectives

Understand top-down firewall rule processing.

Create narrowly scoped firewall rules.

Use protocol, source, destination and port fields correctly.

Identify blocked traffic in OPNsense Live View.

Correlate firewall decisions with packet-capture evidence.

Restore normal connectivity after testing.

3. Baseline Connectivity – E1

Test	Observed result	Status
Ping	4 packets received, 0% packet loss	PASS
DNS	example.com resolved successfully	PASS
HTTP	Successful response headers	PASS
HTTPS	Successful response headers	PASS

Caption: Successful baseline ping, DNS, HTTP and HTTPS tests.

4. Firewall Policy Configuration – E2

Two temporary rules were placed above the broad Default allow LAN to any rule because OPNsense evaluates interface rules from top to bottom.

Rule	Protocol	Destination	Purpose
LAB2 BLOCK ICMP TO 1.1.1.1	ICMP	1.1.1.1	Block selected ICMP traffic
LAB2 BLOCK OUTBOUND HTTP	TCP/80	Any	Block unencrypted HTTP

Caption: LAN rules showing both Lab 2 block rules above the default allow rule.

5. ICMP Test – E3

After the ICMP rule was applied, the Ubuntu client received no ICMP replies from the selected test address. DNS resolution and HTTPS remained available, demonstrating protocol- and destination-specific filtering.

Caption: Failed ICMP test with successful DNS/HTTPS evidence.

6. HTTP Test – E4

The HTTP request to TCP port 80 failed or timed out, while the HTTPS request to TCP port 443 returned successful response headers. This confirmed that the HTTP restriction did not block permitted HTTPS traffic.

Caption: Failed HTTP test and successful HTTPS test.

7. OPNsense Logs – E5

OPNsense Live View recorded blocked LAN traffic generated by the Ubuntu client. The entries identified the interface, source, destination, protocol, action and matching Lab 2 rule label.

Caption: OPNsense Live View entries for blocked ICMP and TCP port 80 traffic.

8. Wireshark Analysis – E6

Wireshark showed ICMP echo requests leaving the Ubuntu client without successful echo replies. Permitted HTTPS traffic showed a completed TCP connection followed by encrypted TLS traffic.

Caption: Wireshark evidence for blocked traffic and permitted HTTPS.

9. States and Automatic NAT – E7

The OPNsense state table was filtered for the Ubuntu client and showed permitted HTTPS connections. Automatic Source NAT remained enabled, translating the private LAN source address for Internet access through the WAN interface.

Caption: State-table evidence for permitted HTTPS and automatic outbound NAT.

10. Restoration – E8

The two temporary Lab 2 rules were disabled rather than deleted. After applying the changes, ping returned 4/4 replies with 0% loss, DNS resolution succeeded, and both HTTP and HTTPS returned successful response headers.

Caption: Final rules page with test rules disabled and restored connectivity.

11. Analysis Questions

38. Why must the specific block rules be placed above the broad allow rule?

OPNsense evaluates interface rules from top to bottom. The specific block must therefore be encountered before the broad allow rule can permit the traffic.

39. Which five packet attributes are most useful when explaining a firewall decision?

Interface/direction, IP protocol, source address, destination address, and destination port where applicable.

40. Why did blocking ICMP not block HTTPS?

The rule matched only ICMP traffic. HTTPS uses TCP, normally on port 443, so it did not match the ICMP rule.

41. What difference did you observe between blocked TCP port 80 and permitted TCP port 443?

Port 80 traffic failed or timed out, with unsuccessful connection attempts visible in the capture. Port 443 completed its TCP connection and carried encrypted TLS traffic.

42. What role does outbound NAT play with a private IPv4 address?

Outbound NAT translates the private LAN source address to a WAN-side address so the client can communicate with external networks through OPNsense.

43. Why restore the original state?

It prevents temporary security-test rules from affecting later exercises and confirms that the laboratory remains controlled and repeatable.

12. Conclusion

The laboratory successfully demonstrated controlled firewall policy testing with OPNsense. Specific ICMP and HTTP traffic was blocked, HTTPS remained available, matching log and packet-capture evidence was obtained, states and automatic NAT were examined, and normal connectivity was restored.

13. Evidence Checklist

E1 – Baseline connectivity

E2 – LAN rule ordering

E3 – Blocked ICMP with DNS/HTTPS available

E4 – Blocked HTTP with HTTPS permitted

E5 – OPNsense Live View

E6 – Wireshark packet analysis

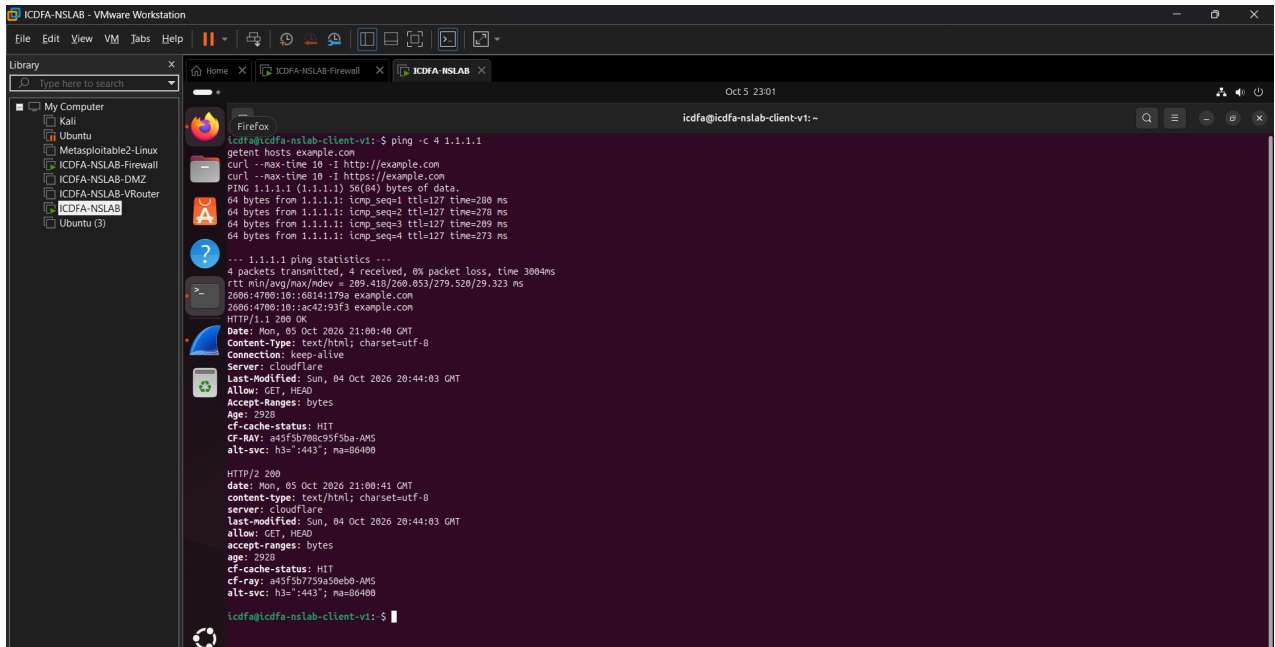
E7 – State-table and automatic NAT

E8 – Final disabled rules and restored connectivity

ACTUAL EVIDENCE SCREENSHOTS

The following screenshots are the actual laboratory evidence captured during the exercise. They have been inserted into the report and labelled E1–E8 in accordance with the laboratory guide.

E1 – Evidence



The screenshot shows a terminal window titled 'icdfa@icdfa-ns-lab-client-v1: ~'. The user has executed several commands to test connectivity:

```
icdfa@icdfa-ns-lab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data:
64 bytes from 1.1.1.1: icmp_seq=1 ttl=127 time=280 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=127 time=278 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=127 time=269 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=127 time=273 ms

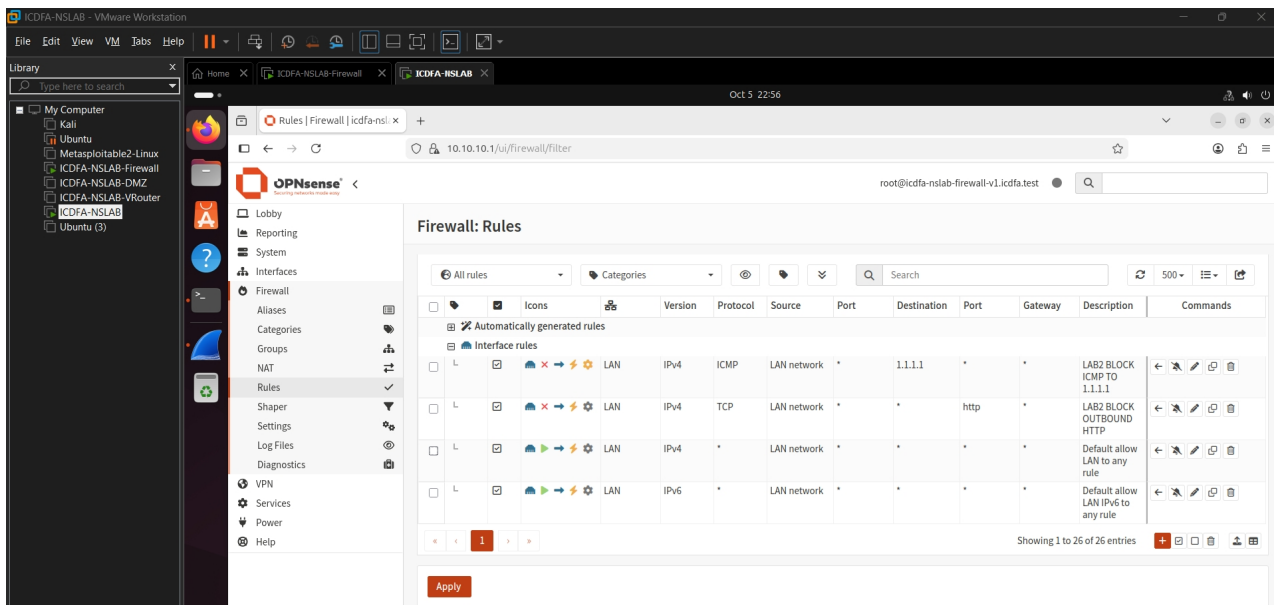
--- 1.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3604ms
rtt min/avg/max/ndev = 269.418/268.053/279.528/29.323 ms
2606:4700:10::6814:179a example.com
2606:4700:10::ac42:93f3 example.com
HTTP/1.1 200 OK
Date: Mon, 05 Oct 2026 21:08:48 GMT
Content-Type: text/html; charset=utf-8
Connection: keep-alive
server: cloudflare
Last-Modified: Sun, 04 Oct 2026 20:44:03 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 2928
cf-cache-status: HIT
cf-ray: a45f5b780c95f5ba-AMS
alt-svc: h3=":443"; ma=86400

HTTP/2 200
date: Mon, 05 Oct 2026 21:08:41 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Sun, 04 Oct 2026 20:44:03 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 2928
cf-cache-status: HIT
cf-ray: a45f5b7759a50eb0-AMS
alt-svc: h3=":443"; ma=86400

icdfa@icdfa-ns-lab-client-v1:~$
```

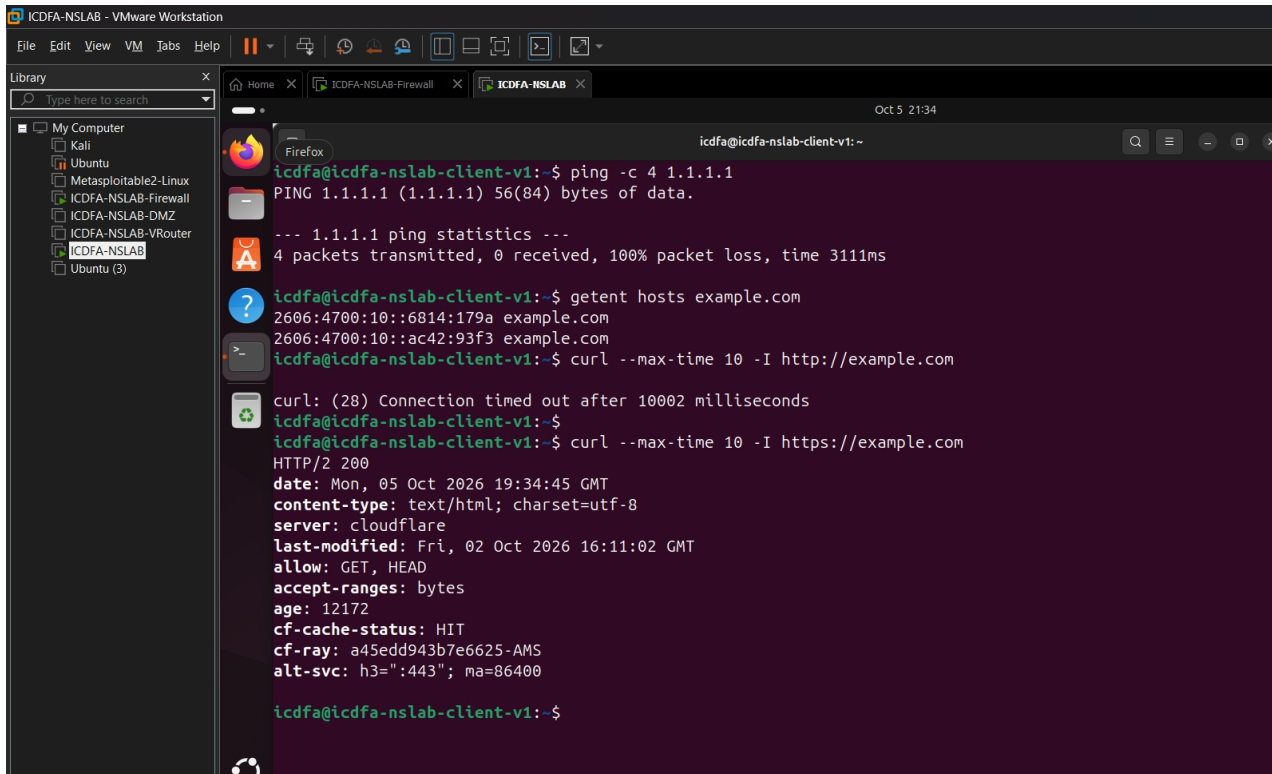
Successful baseline connectivity: ping returned 4/4 replies with 0% packet loss, DNS resolved example.com, and HTTP/HTTPS returned successful response headers.

E2 – Evidence



OPNsense LAN rules showing the two Lab 2 blocking rules positioned above the broad Default allow LAN to any rule.

E3 – Evidence



```
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3111ms

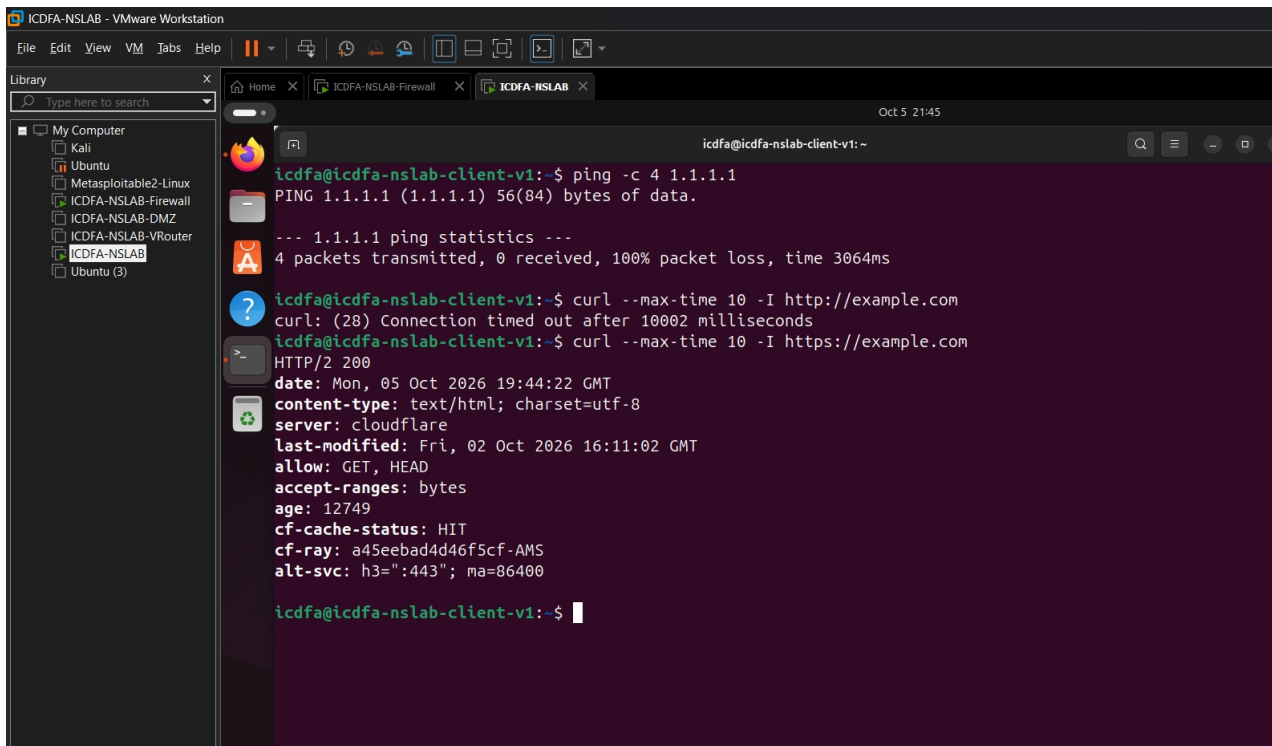
icdfa@icdfa-nslab-client-v1:~$ getent hosts example.com
2606:4700:10::6814:179a example.com
2606:4700:10::ac42:93f3 example.com
icdfa@icdfa-nslab-client-v1:~$ curl --max-time 10 -I http://example.com

curl: (28) Connection timed out after 10002 milliseconds
icdfa@icdfa-nslab-client-v1:~$ curl --max-time 10 -I https://example.com
HTTP/2 200
date: Mon, 05 Oct 2026 19:34:45 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Fri, 02 Oct 2026 16:11:02 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 12172
cf-cache-status: HIT
cf-ray: a45edd943b7e6625-AMS
alt-svc: h3=":443"; ma=86400

icdfa@icdfa-nslab-client-v1:~$
```

The ICMP test produced 100% packet loss while DNS resolution and HTTPS remained available.

E4 – Evidence



```
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3064ms

icdfa@icdfa-nslab-client-v1:~$ curl --max-time 10 -I http://example.com
curl: (28) Connection timed out after 10002 milliseconds
icdfa@icdfa-nslab-client-v1:~$ curl --max-time 10 -I https://example.com
HTTP/2 200
date: Mon, 05 Oct 2026 19:44:22 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Fri, 02 Oct 2026 16:11:02 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 12749
cf-cache-status: HIT
cf-ray: a45eebad4d46f5cf-AMS
alt-svc: h3=":443"; ma=86400

icdfa@icdfa-nslab-client-v1:~$
```

The outbound HTTP request timed out while the HTTPS request returned successful response headers.

E5 – Evidence

The screenshot shows the OPNsense Firewall Log Files Live View interface. The left sidebar contains a navigation menu with options like Lobby, Reporting, System, Interfaces, Firewall, Aliases, Categories, Groups, NAT, Rules, Shaper, Settings, Log Files, General, Live View, Overview, Plain View, and Diagnostics. The main panel displays a table of log entries. The table has columns for Interface, Time, Protocol, Source, Destination, Action, and Label. The log entries show blocked ICMP traffic from 10.10.10.2 to 1.1.1.1, labeled 'LAB2 BLOCK ICMP TO 1.1.1.1'. The table also shows pass traffic for various destinations, labeled 'let out anything from firewall host itself (force gw)'.

Interface	Time	Protocol	Source	Destination	Action	Label	
LAN	In	2026-10-05T20:55:19	ICMP	10.10.10.2	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
LAN	In	2026-10-05T20:55:18	ICMP	10.10.10.2	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
LAN	In	2026-10-05T20:55:17	ICMP	10.10.10.2	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
LAN	In	2026-10-05T20:55:16	ICMP	10.10.10.2	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
WAN	Out	2026-10-05T20:54:59	UDP	10.250.0.128:123	162.159.200.1:123	pass	let out anything from firewall host itself (force gw)
WAN	Out	2026-10-05T20:54:57	UDP	10.250.0.128:123	45.139.163.87:123	pass	let out anything from firewall host itself (force gw)
WAN	Out	2026-10-05T20:54:57	UDP	10.250.0.128:123	192.87.36.4:123	pass	let out anything from firewall host itself (force gw)
WAN	Out	2026-10-05T20:54:57	UDP	10.250.0.128:123	162.159.200.123:123	pass	let out anything from firewall host itself (force gw)
WAN	Out	2026-10-05T20:54:56	UDP	10.250.0.128:123	178.239.19.62:123	pass	let out anything from firewall host itself (force gw)
WAN	Out	2026-10-05T20:54:54	UDP	10.250.0.128:123	103.149.168.181:123	pass	let out anything from firewall host itself (force gw)
WAN	Out	2026-10-05T20:54:54	UDP	10.250.0.128:123	176.97.192.150:123	pass	let out anything from firewall host itself (force gw)

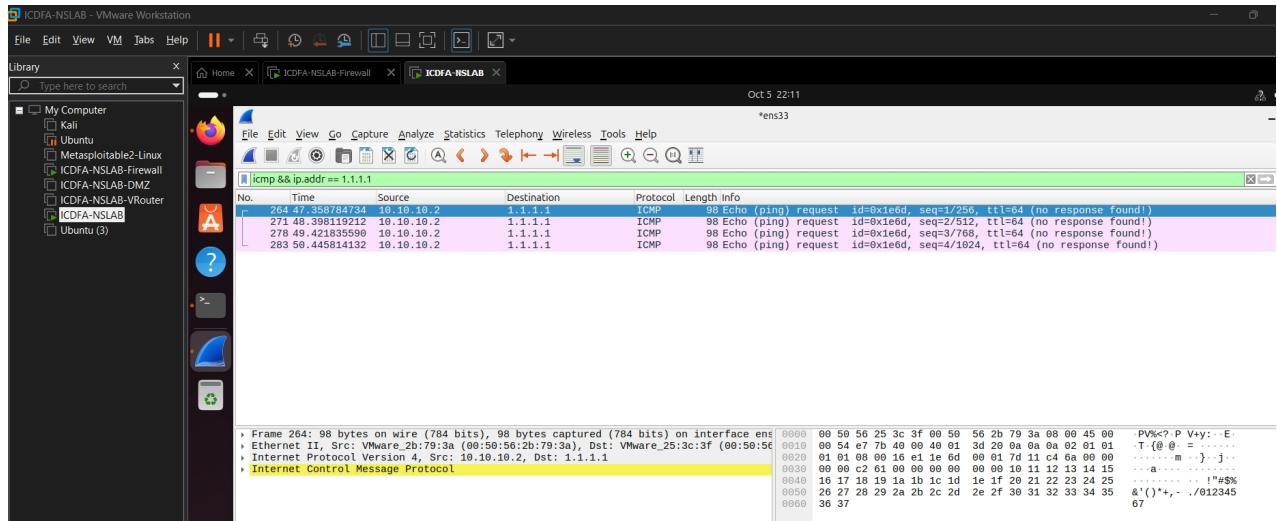
OPNsense Live View records the blocked ICMP traffic and blocked TCP port 80 traffic under the corresponding Lab 2 rule labels. (Evidence image 1 of 2.)

The screenshot shows the OPNsense Firewall Log Files Live View interface. The left sidebar contains a navigation menu with options like Lobby, Reporting, System, Interfaces, Firewall, Aliases, Categories, Groups, NAT, Rules, Shaper, Settings, Log Files, General, Live View, Overview, Plain View, and Diagnostics. The main panel displays a table of log entries. The table has columns for Interface, Time, Protocol, Source, Destination, Action, and Label. The log entries show blocked TCP traffic from 10.10.10.2:38026 to 185.125.190.100:80, labeled 'LAB2 BLOCK OUTBOUND HTTP'. The table also shows pass traffic for various destinations, labeled 'let out anything from firewall host itself (force gw)'.

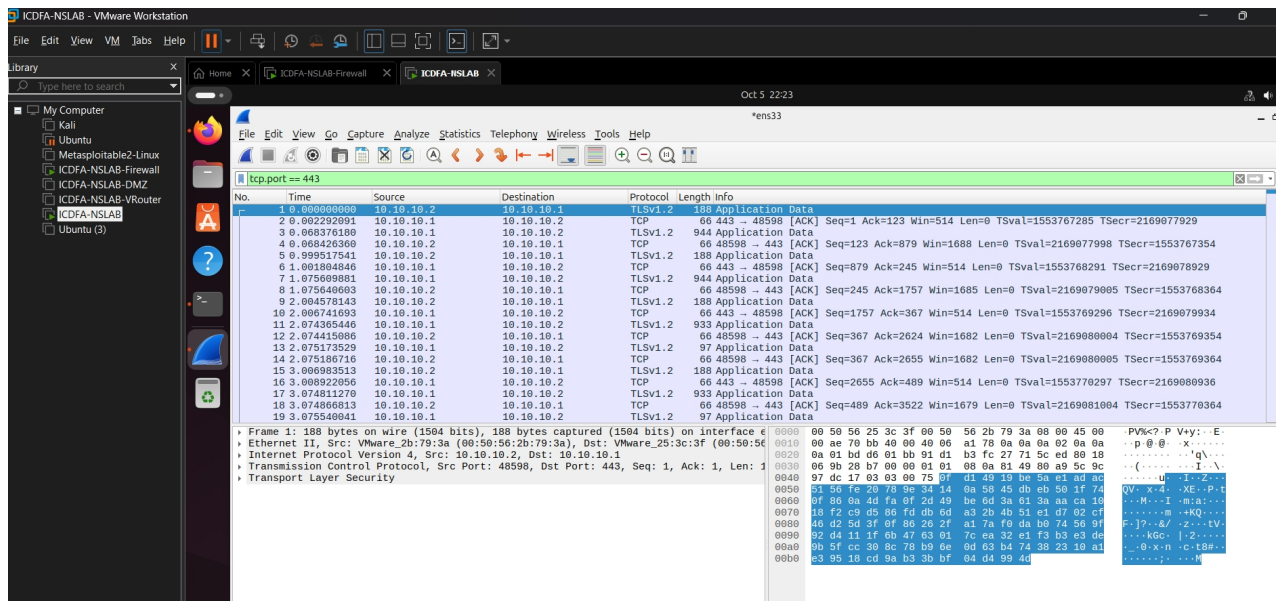
Interface	Time	Protocol	Source	Destination	Action	Label	
LAN	In	2026-10-05T20:51:55	TCP	10.10.10.2:38026	185.125.190.100:80	block	LAB2 BLOCK OUTBOUND HTTP
LAN	In	2026-10-05T20:51:47	TCP	10.10.10.2:38026	185.125.190.100:80	block	LAB2 BLOCK OUTBOUND HTTP
LAN	In	2026-10-05T20:51:42	TCP	10.10.10.2:38026	185.125.190.100:80	block	LAB2 BLOCK OUTBOUND HTTP
LAN	In	2026-10-05T20:51:41	TCP	10.10.10.2:38026	185.125.190.100:80	block	LAB2 BLOCK OUTBOUND HTTP
LAN	In	2026-10-05T20:51:40	TCP	10.10.10.2:38026	185.125.190.100:80	block	LAB2 BLOCK OUTBOUND HTTP
LAN	In	2026-10-05T20:51:39	TCP	10.10.10.2:38026	185.125.190.100:80	block	LAB2 BLOCK OUTBOUND HTTP
LAN	In	2026-10-05T20:51:37	TCP	10.10.10.2:38026	185.125.190.100:80	block	LAB2 BLOCK OUTBOUND HTTP
LAN	In	2026-10-05T20:51:36	TCP	10.10.10.2:38026	185.125.190.100:80	block	LAB2 BLOCK OUTBOUND HTTP
LAN	In	2026-10-05T20:51:35	TCP	10.10.10.2:38026	185.125.190.100:80	block	LAB2 BLOCK OUTBOUND HTTP
WAN	Out	2026-10-05T20:51:35	UDP	10.250.0.128:4672	1.1.1.1:53	pass	let out anything from firewall host itself (force gw)
WAN	Out	2026-10-05T20:51:33	UDP	10.250.0.128:123	45.139.163.87:123	pass	let out anything from firewall host itself (force gw)

OPNsense Live View records the blocked ICMP traffic and blocked TCP port 80 traffic under the corresponding Lab 2 rule labels. (Evidence image 2 of 2.)

E6 – Evidence

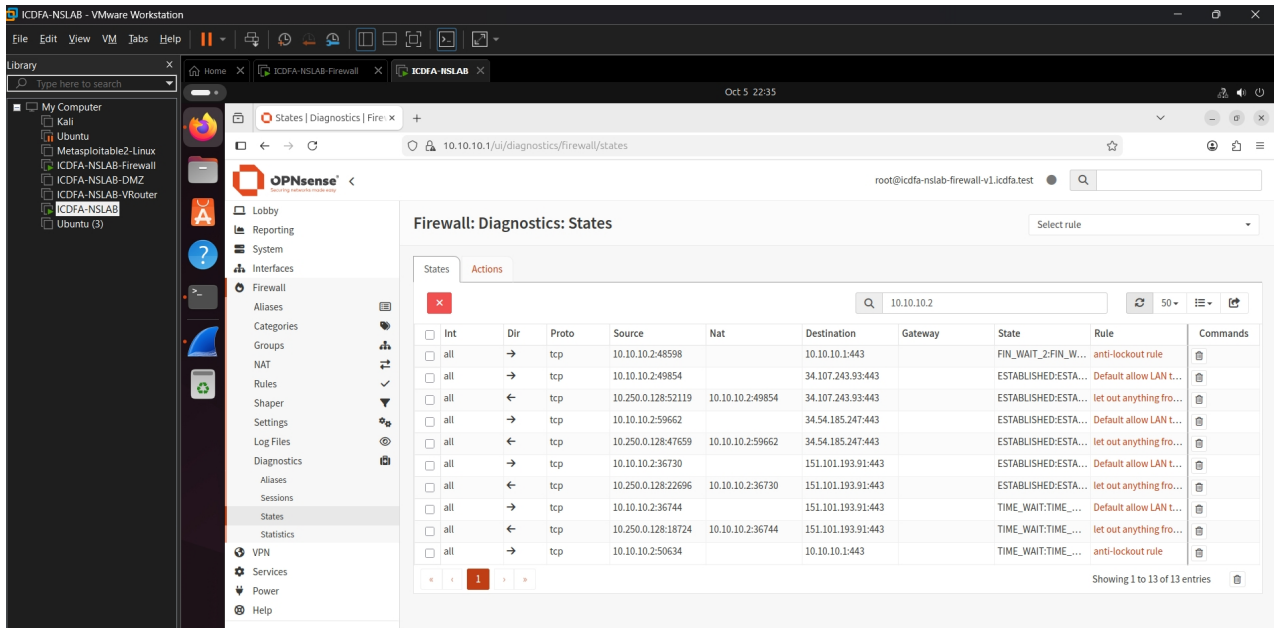


Wireshark shows ICMP echo requests without replies and permitted TCP/443 traffic with an established encrypted TLS exchange. (Evidence image 1 of 2.)



Wireshark shows ICMP echo requests without replies and permitted TCP/443 traffic with an established encrypted TLS exchange. (Evidence image 2 of 2.)

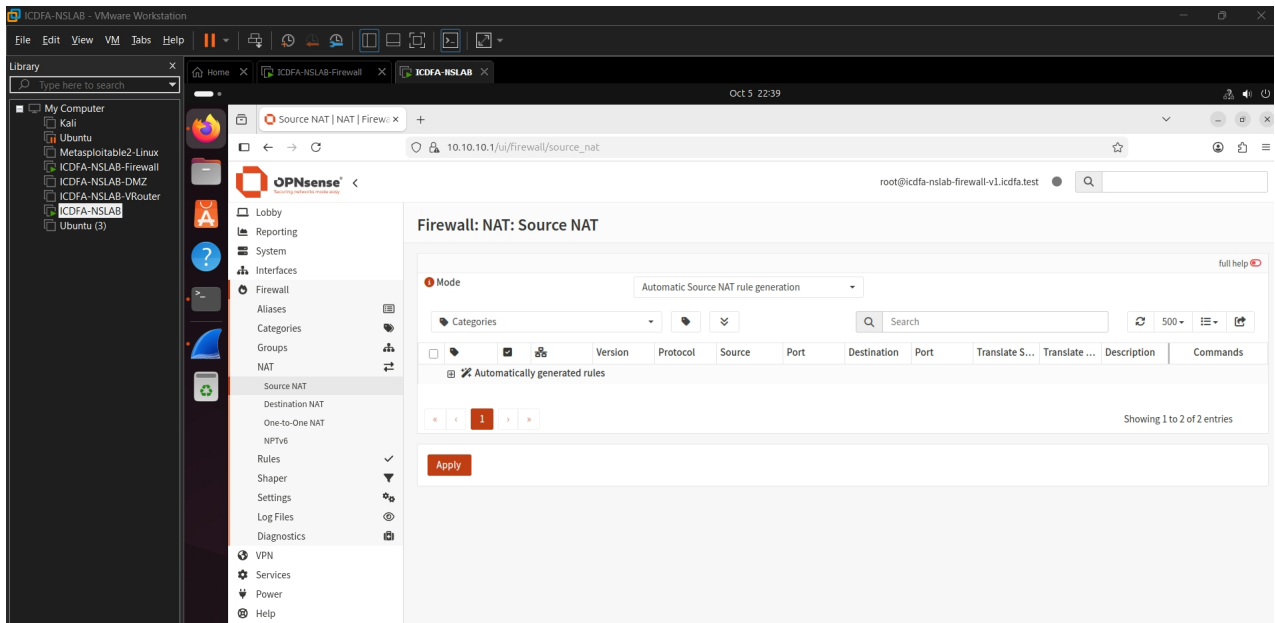
E7 – Evidence



The screenshot shows the OPNsense web interface in a browser window. The left sidebar contains a menu with categories like Firewall, NAT, Rules, and Settings. The main content area is titled "Firewall: Diagnostics: States". It features a search bar and a table of active connections. The table has columns for Int, Dir, Proto, Source, Nat, Destination, Gateway, State, Rule, and Commands. The first row shows a connection from 10.10.10.2 to 10.10.10.1443 on port 443, with the rule "anti-lockout rule".

Int	Dir	Proto	Source	Nat	Destination	Gateway	State	Rule	Commands
all	→	tcp	10.10.10.2:48598		10.10.10.1443		FIN_WAIT_2:FIN_W...	anti-lockout rule	
all	→	tcp	10.10.10.2:49854		34.107.243.93:443		ESTABLISHED:ESTA...	Default allow LAN L...	
all	←	tcp	10.250.0.128:52119	10.10.10.2:49854	34.107.243.93:443		ESTABLISHED:ESTA...	let out anything fro...	
all	→	tcp	10.10.10.2:59662		34.54.185.247:443		ESTABLISHED:ESTA...	Default allow LAN L...	
all	←	tcp	10.250.0.128:47659	10.10.10.2:59662	34.54.185.247:443		ESTABLISHED:ESTA...	let out anything fro...	
all	→	tcp	10.10.10.2:36730		151.101.193.91:443		ESTABLISHED:ESTA...	let out anything fro...	
all	←	tcp	10.250.0.128:22696	10.10.10.2:36730	151.101.193.91:443		ESTABLISHED:ESTA...	let out anything fro...	
all	→	tcp	10.10.10.2:36744		151.101.193.91:443		TIME_WAIT:TIME...	Default allow LAN L...	
all	←	tcp	10.250.0.128:18724	10.10.10.2:36744	151.101.193.91:443		TIME_WAIT:TIME...	let out anything fro...	
all	→	tcp	10.10.10.2:50634		10.10.10.1443		TIME_WAIT:TIME...	anti-lockout rule	

OPNsense state-table evidence shows permitted HTTPS connections for the client; automatic Source NAT is shown separately. (Evidence image 1 of 2.)

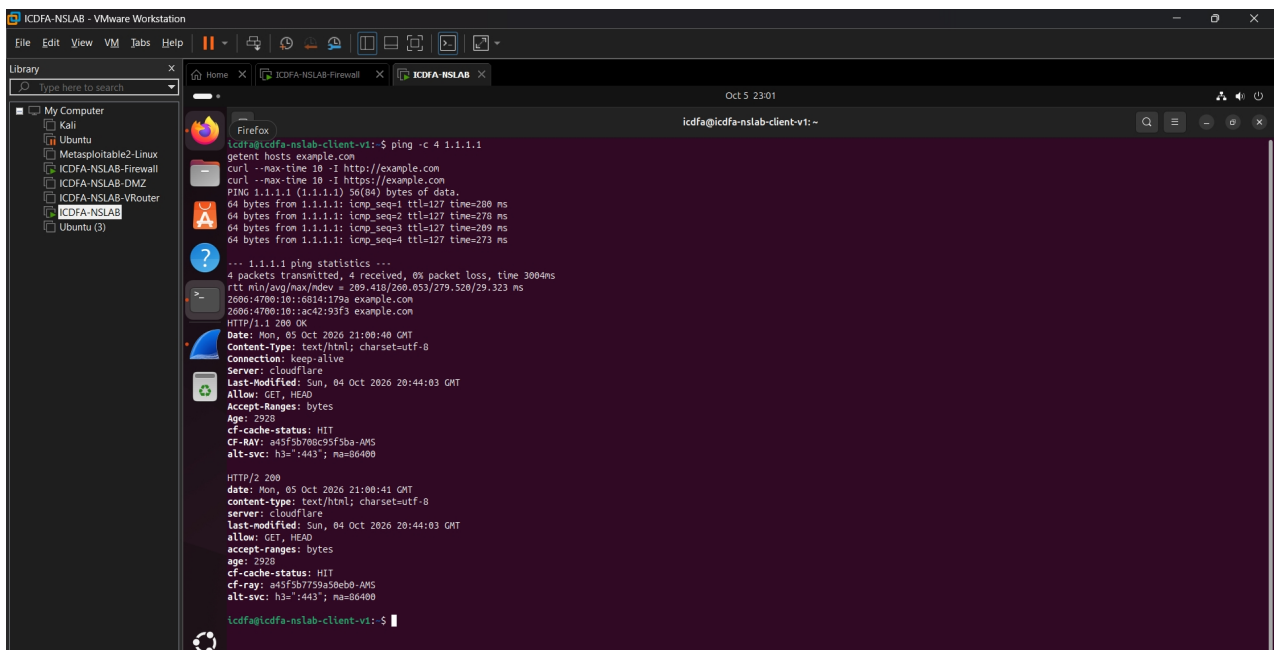
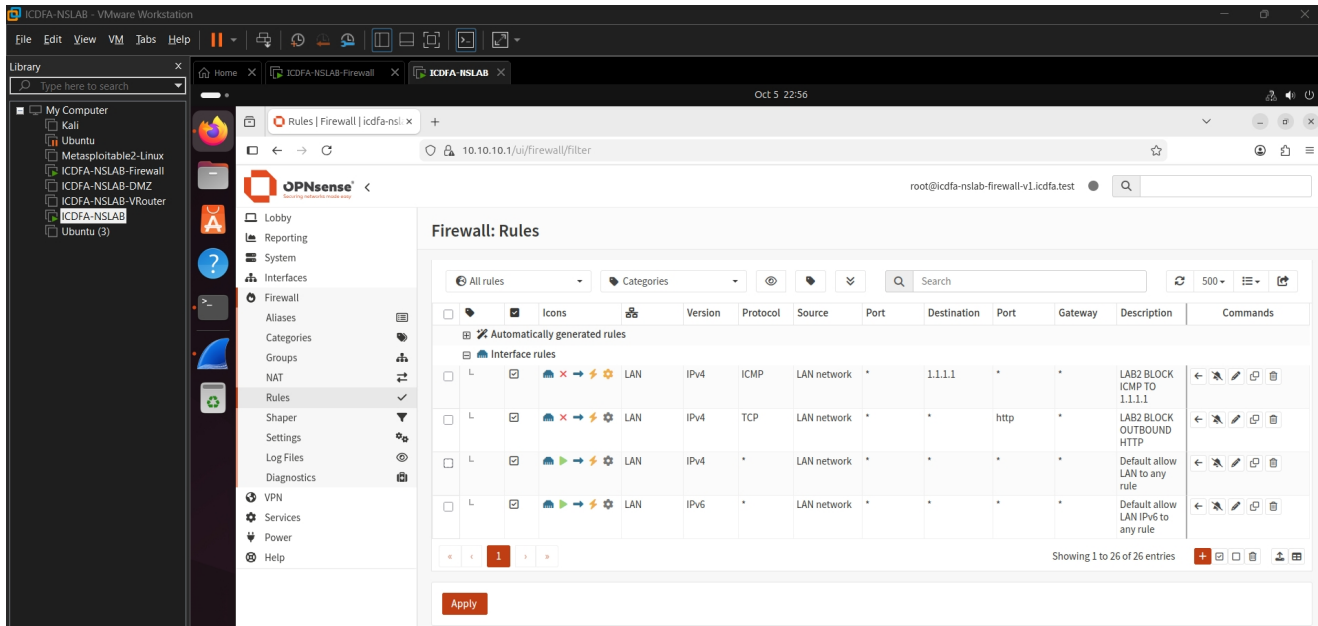


The screenshot shows the OPNsense web interface in a browser window. The left sidebar contains a menu with categories like Firewall, NAT, Rules, and Settings. The main content area is titled "Firewall: NAT: Source NAT". It features a search bar and a table of automatically generated rules. The table has columns for Version, Protocol, Source, Port, Destination, Port, Translate S..., Translate ..., Description, and Commands. The first row shows a rule for "Automatically generated rules" with the description "Automatically generated rules".

Version	Protocol	Source	Port	Destination	Port	Translate S...	Translate ...	Description	Commands
								Automatically generated rules	

OPNsense state-table evidence shows permitted HTTPS connections for the client; automatic Source NAT is shown separately. (Evidence image 2 of 2.)

E8 – Evidence



Restored connectivity after the temporary policy tests: ping, DNS, HTTP and HTTPS were successful. A separate final rules screenshot with both test rules visibly disabled was not among the available screenshots.